PRINCIPLES OF FISH BIOLOGY (ZOO603P)

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SHORT QUESTIONS(3)

1. Set line

In this line several books are used in a single line. The set line consists of mainline, branch line, snood wire and hooks. The branch line gets attached with main line at regular interval.

2. Why we classify Fishes?

We classify fishes to identify and distinguish between species, organize them logically, and facilitate communication among scientists, researchers, and fishermen. Classifying fishes helps in understanding their biodiversity, behavior, evolutionary relationships, and ecological roles, which is essential for effective conservation and management of aquatic ecosystems. It aids in conservation, fisheries management, and scientific research. ***For example***, classifying sharks into different species helps identify which are endangered and need protection measures.

3. Write down the measurements of fishes related to taxonomy

1) Determine family using a key
2) Identify to the lowest taxonomic unit
3) Verify with illustrations
4) Match with previously identified specimens
5) Confirm geographical range
6) Compare with standard texts (FAO, Smith's, Catalog of Fishes, Fishbase)

4. Methods of Measurement of Fish

Fish measurement involves various methods to ensure accurate data collection for scientific and commercial purposes. Common methods include:

1) ***Total Length (TL):*** Measurement from the tip of the snout to the end of the longest caudal fin ray
2) ***Standard Length (SL):*** Measurement from the tip of the snout to the base of the caudal fin, excluding the fin rays
3) ***Fork Length (FL):*** Measurement from the tip of the snout to the fork in the tail
4) ***Weight:*** Using a balance or scale to measure the fish's weight

5. Descriptive Measures of Fish Identification

Descriptive measures of fish identification involve morphological and meristic characteristics:

Morphological: Morphological characteristics provide information about a fish's overall appearance. Body shape, fin placement, scale patterns, and coloration etc.

Meristic: Meristic characteristics involve counting specific features to distinguish between species. Countable features like fin rays, scales along the lateral line, and gill rakers.

6. Meristic Count Related to Taxonomy

Meristic count is a taxonomic tool used to identify and classify fish species based on countable characteristics such as fin rays, scales, and gill rakers. It involves counting the number of features like dorsal fin rays, anal fin rays, pectoral fin rays, scales along the lateral line, and gill rakers to create a unique numerical profile for each species. This profile helps taxonomists distinguish between similar-looking species, understand evolutionary relationships, and group species into genera and families. Meristic count is a crucial aspect of ichthyological research, providing valuable information for fish classification, systematics, and biodiversity studies.

7. Which character include in fish morphology/Parameters for identification of Fish

Specific characters help identify and classify fish species, and understand their evolution, adaptation, and diversity. These are;

- Body shape
- Mouth type

- Mouth location
- Scales
- Gills
- Fin type
- Barbels
- Lateral line system

8. Descriptive Measures of Identification of *Labeo rohita?*/ Parameters of *Labeo rohita* (Rohu fish)

- ***Morphology/Body Shape:*** Elongated and slightly compressed body with a depth contour.
- ***Size:*** Up to 1 m (3.3 ft) in length, 45 kg (99 Ibs) in weight
- ***Coloration***: Silver body with a bluish tinge on the dorsal side, with a distinctive pattern of spots or markings.
- ***Fins***: Dorsal fin with 12-16 rays, pectoral fins with 16-18 rays, and a forked caudal fin.
- ***Scales***: Large cycloid scales covering the body.
- ***Snout***: Short, pointed
- ***Mouth:*** Inferior,
- ***Pharyngeal teeth:*** 3 rows, 5-6 teeth each
- ***Reproductive System:*** Ovoviviparous, with a pair of gonads, and a distinctive structure of reproductive organs in males and females
- ***Digestive System:*** A long alimentary canal with a modified stomach and intestine, with a length ratio of approximately 1:3.

9. Fixation Process

☐ Specimen should be fixed immediately after collection.

☐ To fix the specimen wide mouthed glass filled with fixative solution is can be used.

☐ Tight lids must be used for fixation.

☐ The specimen must be preserved in as natural as possible.

☐ The fixative should be injected to the body cavity to facilitate penetration and preservation of internal organs.

10. Why we choose right side of abdomen for operation in fishes ?

Make a small incision on the right side of the abdomen to facilitate penetration of the preservative. The right side is chosen for this and similar operations like the removal of scale samples because the left side is used for getting morphometric data

11. Physical Parameters of Water Quality

Physical parameters of water quality are those that are determined by the senses of sight, smell, taste, and touch. These physical parameters include temperature, color, taste and odor, turbidity, and content of dissolved solids.

12. Taxonomic Importance of Fish in Museum

Taxonomy and Classification:

Fish museums are crucial for the study of fish taxonomy and classification. They help in the identification and description of new species and in clarifying the evolutionary relationships among different fish groups. Accurate taxonomy is fundamental to understanding biodiversity and conservation.

13. Sensory regulation in fish related to eyes and lateral lines.

Fish rely on two key sensory systems for survival: their eyes and lateral lines. Their eyes detect light, color, and polarized light, guiding navigation, foraging, and communication. The lateral line system, comprising sensors along the body and skin, detects water vibrations, movements, and pressure changes, allowing fish to sense prey, predators, and their surroundings. Together, these systems provide a comprehensive understanding of their environment, enabling fish to respond to threats, find food, and interact with their ecosystem.

14. Rohu fish lives in depths of water when you dissect the fish which organ is responsible for the swimming of rohu fish in depths of water

Swim Bladder: Rohu possesses a swim bladder, an internal gas-filled organ that helps it regulate buoyancy in the water. By adjusting the amount of gas in the swim bladder, rohu can control its depth in the water column.

15. What is the role of gill arches and gill rakers in the fish respiration and feeding process

Gill arches and gill rakers play crucial roles in both respiration and feeding in fish. During respiration, gill arches support the gills, providing a framework for gas exchange, and help pump water over the gills, facilitating oxygen absorption. Gill rakers protect the gills from large particles and debris. In feeding, gill arches funnel food towards the pharynx, while gill rakers filter small food particles from the water, trapping them for swallowing, and prevent large particles from entering the pharynx, protecting the gills. Overall, gill arches provide structural support and facilitate water flow for respiration, while gill rakers filter food particles and protect the gills during feeding, allowing fish to efficiently breathe and feed.

16. Spines and Soft Rays Difference

Fin-rays are fine-supporting structures. Fin rays are the bony or cartilaginous structures that support the fins of a fish. They are of two types:

1) ***Spines (single ray):*** Spines in the fins of Labeo are unsegmented, uniserial structures. Hard and usually sharp structures that provide defense.
2) ***Soft Rays (segmented rays):*** Soft rays are segmented, often branched and biserial. Soft rays are flexible, helping with swimming, maneuverability, and stability.

17. Pelvic Girdle and Pelvic Fin in Rohu

☐ Pelvic girdle is composed of two pelvic bones. Pelvic fin is composed of nine fin rays and three radials.

☐ Fin rays give support to pelvic fin.

☐ Articulation of fin is performed by using radials

18. Components of Pelvic Girdle and Pelvic Fin

Pelvic Girdle:

Pelvic girdle is composed of two pelvic bones.

- ischium and
- the pubic bone.

Pelvic Fin:

- nine fin rays and
- three radials.

19. Cranium of *Labeo rohita*.

- ***Orbital Regions:*** It is composed of Supra-orbital, Post-frontal, Pre-orbital, Infra-orbital and Post orbital.
- ***Ethmoidal or Nasal Region:*** It is composed of Nasal, Ecto-ethmoid, Lacrymals, Mesethmoid, vomer and Rostral bones.
- ***Temporal Region:*** It consists of Parietal, Alisphenoids, Parasphenoids and Frontal bones.
- ***Occipital region:*** It is composed of bones: Basi-occipital, Ex-occipital and supra occipital.
- ***Auditory or Otic region:*** It consists of bones: Pro-otic, Epi-otic, Sphenotic and Pterotic.

LONG QUESTIONS(5)

1. Define HEL, PEL and PVL.

Head Length (HEL): Taken from the tip of the snout to the posterior most point reached by the bony margin of the operculum

Pectoral fin length (PEL): Distance from the extreme base of the uppermost ray to the farthest tip of the fin, filament if any.

Pelvic fin length (PVL): Distance from the extreme base of the uppermost ray to the farthest tip of the fin, filament if any.

2. Classification of Labeo

Labeo rohita, commonly known as Rohu, is a species of fish in the carp family. Here is its taxonomic classification:

- **Kingdom** : Animalia
- **Phylum** : Chordata
- **Class** : Actinopterygii
- **Order** : Cypriniformes
- **Family** : Cyprinidae

- **Genus** : Labeo
- **Specie** : *Labeo rohita*

3. Enlist the bones of *Labeo rohita's* skull / Cranium of *Labeo rohita.*

Occipital region: It is composed of bones: Basi-occipital, Ex-occipital and supra occipital.

- ***Auditory or Otic region:*** It consisits of bones: Pro-otic, Epi-otic, Sphenotic and Pterotic.
- ***Temporal Region:*** It consists of Parietal, Alisphenoids, Parashenoids and Frontal bones.
- ***Orbital Regions:*** It is composed of Supra-orbital, Post-frontal, Pre-orbital, Infra-orbital and Post orbital.
- ***Ethmoidal or Nasal Region:*** It is composed of Nasal, Ecto-ethmoid, Lacrymals, Mesethmoid, vomer and Rostral bones.
- ***Splanchnocranium (Viscerocranium):*** Premaxilla, Maxilla, Dentary, Palatine, Opercular bones (including operculum, preoperculum, suboperculum, and interoperculum)
- ***Branchial Arches:*** Consist of Epibranchials, Ceratobranchials, Hypobranchials, and Basibranchials.

4. How many vertebrae are in *Labeo rohita* and what it's role

The vertebral column of Rohu fish consists of 37-38 vertebrae. The vertebrae are bony and amphicoelous in nature (concave at both ends of the centrum). The vertebral column is divided into two regions. There are 21 vertebrae in the front part of the trunk region. And there are 16-17 caudal vertebrae in the posterior part of the tail region.

From the first vertebral neural arch, two small bones originate, called Claustrum and Scaphium. They are the most frontal part of the Weberian ossicles. The second and third vertebral centrum are joined together to form a flattened triangular bone on each side, the outer side of which is like a slice of the moon. It is called Tripus. There is a small bone called the intercalarium on each side between the scapula and the tripus. The scaphium edge of this chain connects to the inner ear and the tripus to the swimbladder, helping the sound waves felt in the swimbladder or air sac to reach the inner ear.

- **IMPORTANCE:**

The vertebrae in *Labeo rohita* are essential for:

- ***Structural Support:*** Maintaining body shape and posture.
- ***Protection***: Shielding the spinal cord.
- ***Movement***: Allowing flexible, coordinated swimming motions via articulation between vertebrae.
- ***Muscle Attachment:*** Providing anchor points for muscles, essential for various movements.
- ***Fin Support:*** Supporting the caudal fin for propulsion and maneuvering.
- ***Buoyancy Control:*** Contributing to overall stability and buoyancy through the structure and rigidity of the vertebral column.

5. Special Adaptive Features of Fish skull

- Occipital process is reduced and it shows lack of neck.
- Orbital and temporal bones are flat.
- Maxilla and Pre-maxilla is ventrally placed and it result in increased entry of water through mouth.
- Nasal bone is closed at the Posterior end which reduces the entry of water in to nasal cavity.
- The skull bone of the fish is flat and this reduces the friction.
- Jaw bones are supported by dentary bone.
- Trinagular bone is quadrate which limits the movement of lower jaw.
- In Labeo the jaw bone is reduced.

6. Formalin fixation steps

The collected specimens should be first fixed in the preservatives (10% formalin or 70% ethanol) in a large container so that the specimen maintains its original shape after fixation. Preservation in tight containers distorts the shape. Fishes may also be Preserved in buffered 10% formalin. ***Buffered formalin*** is prepared by adding three grams of borax per litre of 10% formalin solution. It retards the tissue shrinkage and prevents decalcification of the tissue. A general rule is maintained not to have more than 40% of biomass per container of formalin during fixation.

7. Preservation of Fish by Using Formalin Solution

- **Preparation:**

Use 10% formalin solution (1 part formaldehyde to 9 parts water) or buffered 10% formalin (with 3g borax per liter) to prevent tissue shrinkage and decalcification.

- **Method:**

Fix the collected specimens in a large container with the preservative, ensuring the specimen maintains its original shape. Avoid overloading the container (max 40% biomass per container). .

After fixation the specimens are put in a glass-stoppered bottles or screw capped jars with the snout pointing downwards and then filled with freshly prepared preservative. An incision should be made on the abdominal wall of the fish length 10-30 cm to one side of the mid-ventral line with a scalpel knife or scissor. And for the fish longer than 30 cm undiluted concentrated formalin is injected with the help of a hypodermic syringe throughout the abdomen.

- **Time:**

The time of the formalin fixation of the average specimen may range from two days to a week.

8. For permanent preservation, formaldehyde will be removed? How? And what is the reason? How to apply formaldehyde while preserving fish and why it is important to remove while permanent preservation of fish.

Formaldehyde is often used as a preservative in biological samples, but it can be toxic and carcinogenic. For permanent preservation, formaldehyde is typically removed and replaced with a safer alternative, such as ethanol or glycerin.

- **Applying formaldehyde for fish preservation:**

Initial fixation: Immerse the fish in a 10% formaldehyde solution for 2-4 weeks to prevent decay and fix tissues.

Concentration adjustment: Gradually increase the formaldehyde concentration to 20-30% for long-term preservation.

- **Process of removing**:

For permanent preservation, formaldehyde can be removed by soaking the specimen at least for two days in freshwater with a regular water change.

- **Permanent Preservation:**

After removal, transfer the specimen to 70% ethanol for long-term storage. Ethanol preserves tissue integrity and prevents microbial growth. The permanent preservation can be done in 70% alcohol or 40 % alcohol.

- **Reason:**

Formaldehyde is toxic and can cause tissue hardening and color loss over time. The reason for removing formaldehyde is to prevent damage to the sample over time, as well as to reduce the risk of exposure to toxic fumes during handling and storage. Additionally, formaldehyde can interfere with downstream applications, such as DNA or protein analysis, making its removal necessary for long-term preservation and research purposes.

9. Factors affecting water quality

Water quality is influenced by anthropogenic activities and natural factors. These are some of the factors which affect water quality.

☐ Atmospheric pollution

☐ Runoff

☐ Erosion and Sedimentation

10. Parameters of water quality

- **Physical parameters of water quality**

Physical parameters of water quality are those that are determined by the senses of sight, smell, taste, and touch. These physical parameters include temperature, color, taste and odor, turbidity, and content of dissolved solids.

- **Chemical parameters of water quality**

Chemical parameters of water quality are measures of those characteristics which reflect the environment with which water has contact. These chemical parameters can measure pH, hardness, amount of dissolved oxygen, biochemical oxygen demand (BOD), chemical oxygen demand (COD), and levels of chloride, chlorine residual, sulfate, nitrogen, fluoride, iron and manganese, copper and zinc, toxic organic and inorganic substances, as well as radioactive substances.

- **Biological parameters of water quality**

Biological parameters of water quality are those measurements that reflect the number of bacteria, algae, viruses, and protozoa that are present in water.

11. Role of fish Museum in biodiversity conservation, scientific research and educational significance of taxonomy?

- **Biodiversity Conservation:**

Fish museums house extensive collections of fish specimens, including rare and endangered species. These collections serve as:

- invaluable references for researchers and conservationists studying biodiversity and monitoring changes in fish populations over time.
- aid in identifying and understanding threatened or declining species, which is crucial for conservation efforts.

- **Scientific Research:**

Fish museums are essential for scientific research in ichthyology. Researchers use these museums to compare and contrast fish species, study their anatomy, behavior, and genetics, and advance our understanding of aquatic ecosystems. This knowledge is essential for managing and conserving fish populations.

- **Education and Outreach:**

• Fish museums often have educational programs. • They provide a unique opportunity to learn about the incredible diversity of fish, their habitats, and their ecological importance. • Educating the public about fish and aquatic ecosystems • fosters awareness and encourages responsible environmental stewardship.

12. Difference b/w Electrical Conductivity & Hardness

- **Electrical conductivity (EC):**

Electrical conductivity indirectly measures the ionic concentration of water by measuring its ability to carry or conduct an electrical current. Higher conductivity means more solids are present in the water.

- **Hardness:**

Hardness is a property of mineralized water, and it measures the concentrations of certain dissolved minerals, particularly calcium and magnesium. □ Biochemical oxygen demand (BOD): Biochemical oxygen demand indirectly measures the degree of microbial contamination, and is primarily used as a measurement of the power of sewage water.

13. While dissection, how we will determine gender of fish like *Labeo rohita?*

During dissection, the gender of fish can be determined by examining the reproductive organs.

Make a ventral incision, cutting along the belly of the fish from the anus to the base of the gills. Gently pull out the internal organs, including the digestive tract, to expose the reproductive organs.

Look for the gonads attached to the dorsal wall of the body cavity.

- ***Males***: A pair of testes, which are small, pale yellow masses located on the ventral side.
- ***Females***: A single, large yellow ovary filled with eggs.

The testes and ovaries are connected by tubes to the urogenital opening found posterior to the anus.

Check for additional features, such as a papilla near the anus in males, or a genital pore near the anus in females. Ovaries may contain visible eggs, and testes may have a distinct sperm sac.

14. Rohu Digestive System

Stomach and Digestive System: Like all fish, rohu has a stomach and an intestine that are responsible for digesting food, breaking down nutrients, and absorbing them into the bloodstream. In Labeo, true stomach is absent. The oesophagus opens into a swollen sac-like anterior part of intestine. This sac-like part is called intestinal bulb in which food is stored. The oesophageal opening into the intestinal bulb is guarded by an oesophageal valve to prevent regurgitation of digested food. Gastric glands are lacking in intestinal bulb. The mucosa of the intestinal bulb contains absorptive and mucous cells. The intestinal bulb is also differentiated into an anterior broader cardiac part which opens the pancreatic and bile ducts, and a posterior narrow pyloric part. Pyloric caeca is absent. The mucosa of the cardiac part forms honey-comb-like folds and that of the pyloric part forms longitudinal folds. The intestinal bulb is followed by a thin-walled, narrow, elongated, and convoluted intestine. Its mucosa forms oblique transverse folds in the anterior region and longitudinal folds in the posterior region. Since Labeo is a herbivorous fish, its intestine is longer in comparison to carnivorous fishes. Intestinal villi and scroll valve are absent in Labeo.

OBJECTIVE

Methods in which fish is caught by diving and gasing with hand is called: *hand picking*

Our body is made of *70* __ percent of water

study of fish is called: *icthyology*

Formalin is used in fish preservation.

Mucpus glands are present in which part of fish body. *skin*

pelvic gridle plays important role in pelvis fin's function: *by providing structural support*

1. Primary function of scales in fish(protection against predator)

2. Gills in cartilaginous fish (5_7 pairs)

3. Method to catch predator fish(troll lines)

4. Turbidity is(cloudiness of water

5. Ph (increase solubility of heavy metals)

6. Function of anal fin(stabilize in water)

7. Common preservatives (formalin)

8. Museum can facilitate student because (they preserve specimen)

Fish detect odor by...? *olfactory organs (nasal sacs)

Heart present in...? ***anteriorly near the gills**

Human body consists of **70%** water .

Which chemistry is used in preservation .

Male and female reproductive parts? ***testes and ovaries***

Which organ help in deep water **swim bladder**

1. In *drift line*, the one end of line gets fastened with floats, buoys and markers (Flag) whereas other end is tied with boat.

2. For permanent preservation, formaldehyde can be removed by soaking the specimen at least for two days in *freshwater with a regular water change.*

3. Morphology, color, scales, reproductive system is included in *description.*

4. Ichthyology museums, play a significant role in *preserving.*

5. Labeo rohita known as *rohu.*

6. Begin making an incision along the *dorsal side* of the fish.

7. It consists of *8 fin rays.*

8. Turbidity refers to the *cloudiness of water.*

9. Hardness of water is due to certain dissolved minerals, particularly *calcium and magnesium.*

🐬 Hardness of water measure the concentration of certain dissolved minerals particularly *calcium* *and* *magnesium*

🐬 Turbidity refers *Cloudness of water*

🐬 Scales are primerly used to *measure the age*

🐬 incision along the *dorsal side*

🐬 *Caudal fin* use to change the direction of movement in water.

🐬Fish oil extract from the *liver*

🐬Swim bladder regulate *buoyancy in the water*

🐬Fish phylum *Chordata*

what is recommended method to remove formaldehyde for permanent preservation **soaking the specimen in freshwater with regular water changes***

(2)which one important for fish to respire **Gill**

(3) excessively high and low PH level in water can___**increase solubility of heavy metals**

(4)which is the purpose of anal fin___**stabilize the fish in water**

(5)main focus of ichthyology museum on____ **preserving specimens**

Fish eyes has nictaiting membrane **thin transparent membrane is present which is called nictating membrane that protects the eye.**

Definition of troll line method **several lines are attached along with hooks on outrigger of the boat. Only predatory fishes get caught by this line. This line gets operated on surface water body. It is the only line which is active in nature.**

EYD MEANS **Eye diameter (EYD)**

Killing of fish is ethical ?**killing of fish is an ethical treatment of a live animal and also serves scientific purpose**

Which organ help in swimming **swim bladder**

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